Project FORESIGHT

End-to-End Enterprise Retail Analytics, Demand Forecasting & Inventory Intelligence Platform

Document Version	Status	Domain	Target Interface
v1.0 (Production Architecture)	Approved & Deployed	Retail Supply Chain & ML	Interactive Streamlit BI

1. Executive Summary & Overview

Project FORESIGHT is an enterprise-grade retail intelligence platform architected to solve the dual challenges of inventory optimization: *overstock holding depreciation* and *stockout-induced revenue leakage*. By ingesting over 17.65 million transaction records across 50 retail store locations and 200 SKUs, FORESIGHT combines advanced intermittent time-series modeling (Croston/Syntetos-Boylan) with gradient-boosted decision trees (LightGBM) and inventory policy algorithms to generate high-accuracy multi-horizon demand forecasts and prescriptive stock replenishment directives.

50	200	10,000	17.65M+	30/60/90d
RETAIL STORES	ACTIVE SKUS	STORE-SKU PAIRS	HISTORICAL RECORDS	FORECAST HORIZONS

2. Business Context & Strategic Objectives

The Overstock Vulnerability

Excess inventory ties up working capital, inflates warehouse holding and handling costs, leads to product obsolescence/spoilage, and forces margin-eroding promotional markdowns.

The Stockout Threat

Inventory depletion directly triggers lost sales, customer dissatisfaction, defection to competitors, reduced On-Time In-Full (OTIF) fulfillment scores, and impaired brand loyalty.

Core Analytical & Operational Objectives:

- **Granular Demand Profiling:** Uncover top-performing store-SKU nodes, regional sales drivers, price elasticity, day-of-week seasonality, and promotional uplift.
- **Intermittent Demand Handling:** Classify products using the Syntetos-Boylan / ADI-CV^2 framework to select specialized models for zero-demand periods.
- **Multi-Horizon Forecasting:** Generate recursive and direct forecasts across operational (30-day), tactical (60-day), and strategic (90-day) planning windows.
- **Prescriptive Inventory Risk Scoring:** Automatically categorize each Store-SKU into risk quadrants (Overstocked, Severe Stockout Risk, Balanced, Inactive/Dead Stock) and produce exact replenishment guidance.

3. Dataset Scale & Structural Profile

The underlying dataset captures granular daily transactions from January 1, 2021 through October 31, 2025 across distributed regional retail nodes:

Dimension	Value	Analytical Impact & Processing Requirements
Store Count	50 Stores	Distributed across diverse geographic zones with distinct customer demographic patterns.
Product Catalog (SKU)	200 Active SKUs	Spans fast-moving consumer goods (FMCG), slow-movers, and high-value intermittent lines.
Store-SKU Combinations	10,000 Nodes	Requires scalable parallelized feature engineering and hierarchical modeling.
Historical Timeline	2021-01-01 to 2025-10-31	~4.8 years of historical coverage; enables learning annual seasonality, trends, and macro events.
Total Observations	17,650,000+ Rows	Requires memory-efficient parquet storage, vectorized polars/pandas pipelines, and LightGBM histogram binning.
Forecast Horizons	30, 60, 90 Days	Granular daily predictions aggregated into weekly and monthly supply chain procurement cycles.

4. End-to-End System Architecture

The FORESIGHT architectural pipeline follows a modular 5-stage lifecycle ensuring robust data governance, reproducible machine learning, and intuitive executive access:

Stage 1 — Data Ingestion & Data Hygiene: Automated schema validation, missing timestamp backfilling with zero-demand records, outlier clipping using IQR/rolling medians, and store calendar alignment.

Stage 2 — Demand Classification & Driver Diagnostics: Computation of Average Demand Interval (ADI) and Squared Coefficient of Variation (CV^2). Segregation of series into Smooth, Erratic, Intermittent, and Lumpy categories. Feature attribution for price changes, weekend effects, and holidays.

Stage 3 — Multi-Tier Machine Learning Forecasting Engine: Hybrid forecasting using Croston's Method / TSB for sparse lines, Exponential Smoothing / AutoARIMA for smooth lines, and a high-performance *LightGBM Regressor* enriched with rolling lag statistics (7d, 14d, 28d), cyclical calendar features, and hierarchical store-category aggregations.

Stage 4 — Inventory Intelligence & Risk Optimization: Translates point forecasts and prediction variance into Reorder Points (ROP), Safety Stock (SS) levels, and Days of Supply (DOS) relative to live store on-hand inventory.

Stage 5 — Streamlit Decision Portal Deployment: Real-time interactive dashboard featuring dynamic store/SKU filtering, risk alerts, scenario planning sandboxes, and exportable purchase order recommendations.

5. Machine Learning & Forecasting Methodology

Feature Engineering Matrix

- **Lag Features:** Lag 1, 7, 14, 21, 28, 30, 60, 90 days
- **Rolling Statistics:** 7d, 14d, 30d rolling mean, std dev, min, max, and zero-demand ratios
- **Calendar & Trend:** Day of week (sin/cos), day of month, month, quarter, holiday indicator
- **Target Encoding:** Historical mean sales by SKU-Store and Category level

Evaluation & Validation Protocol

- **Temporal Cross-Validation:** Expanding window backtesting with no lookahead bias
- **Weighted Absolute Percentage Error (WAPE):** Primary metric to prevent zero-division distortion
- **Root Mean Squared Scaled Error (RMSSE):** Evaluates volatility penalty on high-volume SKUs
- **Pinball Loss:** Evaluates P10, P50, and P90 quantile estimates for safety stock buffering

6. Inventory Intelligence & Prescriptive Decision Matrix

FORESIGHT evaluates live store stock against projected 30-day cumulative demand and supplier lead times:

Risk Category	Condition (Days of Supply)	Operational Status	System Recommendation
CRITICAL STOCKOUT	Current Stock < Lead Time Demand (DOS < 7d)	Imminent loss of sales & stock depletion	**Expedite Purchase Order**; trigger emergency cross-dock re-allocation.
REORDER TRIGGERED	Stock ≤ Reorder Point (7d ≤ DOS ≤ 21d)	Approaching safety threshold	**Standard Replenishment**: Order Economic Order Quantity (EOQ).
OPTIMAL BALANCE	Healthy buffer (22d ≤ DOS ≤ 45d)	Fulfillment equilibrium	**Maintain Policy**: No immediate procurement action required.
OVERSTOCK WARNING	Excess holding (DOS > 60d)	Capital locked, inventory aging	**Halt Inbound Orders**; rebalance stock to high-demand regional stores.

7. Technology Stack & Enterprise Tooling

Layer	Technologies	Core Purpose
Core Language & Runtime	Python 3.11+, NumPy, SciPy	High-performance numeric computing & pipeline orchestration.
Data Processing & ETL	Pandas, Polars, PyArrow, DuckDB	Scalable columnar manipulation for 17.65M+ records.
Predictive Modeling	LightGBM, Statsmodels, Scikit-Learn	

Layer	Technologies	Core Purpose
		Gradient boosting, intermittent Croston modeling & feature stores.
Visualization & App	Streamlit, Plotly Express, Altair	Dynamic web dashboard, interactive filters & scenario simulators.
Packaging & Output	WeasyPrint, ReportLab, Docker	Automated PDF reporting, containerized cloud microservices.

8. Implementation Roadmap & Value Realization

Expected Business ROI & Impact:

- **18% – 25% Reduction in Safety Stock Holding Costs:** By replacing static buffer rules with dynamic quantile forecasts.
- **30% – 45% Drop in Out-of-Stock Incidents:** Proactive 30-day early warnings prevent supplier lead-time bottlenecks.
- **Automated Buyer Workflow:** Reduces manual purchase requisition preparation time from 14 hours/week to under 30 minutes.

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